

A Packing of 16 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.559634. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.549645, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.559634 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 16$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.549645, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 16/15 = 2.560000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N16.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
341284597023560935224182489292759059120346975453015762721492410647936523
539016680999033032319645601783343938967717696660425304359765271354269512
73810980399085775236247055691626727801010321646
```

denominator:

```
133333333333396227154589427736178316392626759331203029113967829961795448
267285425813572667632085975121719734439912320744640480279313228210512151
09725152651475764622258951742743166415326357981
```

B Coordinates

16 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N16.txt`.

```
371580165536 189168251742 -768655211553 -266052353042342779366 535175971792 -6034620174 4500242884 293400408904 -200419267589 -32920788257 -375059909715
-330530405104 -32795829444 -219603182564 5432486302321136774570 -533162934134 256012490608 8719171781172495296328 256265484702 -447591741170 -78607819920
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